Divij Motwani

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Education

University of California, Berkeley Berkeley, CA
B.S. Electrical Engineering & Computer Sciences *Expected May 2028*

Research Experience

UC Santa Cruz — IR & Knowledge Management Lab Santa Cruz, CA
Student Researcher, Accepted NeurIPS 2025 *Sep 2023 – August 2025*

- Co-authored **RAGUARD** fact-checking benchmark (Accepted NeurIPS 2025 (Main Conference), FEVER-8 2025) spanning 2,648 U.S. presidential-campaign claims (2000–2024) linked to 16,331 Reddit-sourced evidence docs labeled supporting / misleading / unrelated (avg. 6.2 docs/claim).
- Benchmarked seven LLM-RAG systems; misleading-only retrieval caused a mean 46.5% relative accuracy drop; GPT-4o most robust, Claude 3.5 strongest zero-context baseline.
- Engineered scalable LLM-guided labeling workflow to auto-tag misleading vs. supporting evidence at scale.
- 10 Citations (Google Scholar)

OralAI — 1st Place, Regeneron ISEF, Top ∼40 of >10,000,000 students Palo Alto, CA
Co-founder, Nonprovisional Patent Pending; in talks with Philips Sonicare *Jan 2023 – Present*

- Designed UV-fluorescence dental diagnostic system with custom 405 nm intra-oral camera and Raspberry Pi.
- Trained image-segmentation models: 98.4% precision (tooth detection), 93.4% (plaque segmentation).
- Deployed full-stack mobile app with real-time inference; cloud backend on AWS Cloud + MongoDB Atlas.

Selected Projects

ZKPBench, ZKP based DePIN verification system — Noir, Solidity, Base *2025*

- Privacy-preserving network benchmarking: verifiers commit Poseidon-hashed challenges; clients generate local Noir zkSNARKs proving correct responses; Solidity verifier checks proofs and mints an ERC-1155 badge with verified exact latency.
- Abuse controls: 5-min epoch batching and token-bucket rate limiting; deployment on Base (Coinbase L2); no raw logs/IPs leave clients while proofs remain publicly verifiable. Proof of concept for compute/DePIN verification mathematically ensuring trust.

FIRST Robotics Team 6036, Top 10 worldwide | *Build & Software Subteams* *2023 – 2025*

- Veteran Build and Software member, focusing on high scale perception systems for control and tracking with fiducials
- Repurposed NASA JPL's **mrcal** calibration systems on multi-threaded multi-coprocessor architecture to implement with custom autonomous control systems for centimeter level positioning in 3D-space relative to operating field
- Developed and prototyped hardware with CAD/CAM for rapid iteration; weight/stress optimization; maintained motor & battery management systems; proficient with advanced power tools for precise fabrication; served on pit crew for real-time repairs

LEO Satellite Network Simulator — CesiumJS, TypeScript, Solidity *2025*

- Software control system for DePIN LEO satellite mesh networks; connectivity with optical inter-satellite laser links; satellites use custom standardized hardware to receive and send data; geostationary as data exit-nodes with constant ground-based laser links
- Star-tracker localization (GPS-independent) to enter network, improving high-speed orbital network access at low cost.
- Interactive CesiumJS dashboard visualizes latency, network-resilience, collision prevention, and network allocation smart contracts.

Zeta — AI Research Discovery System *AI Agent Hackathon (Exa × Anthropic × AWS × Lightspeed) 2025*

- Crawled and parsed 6,117 latest *arXiv* Computation & Language papers (1.2 M sentences); Claude Opus classifier flagged 32K possible future-work statements and unnoticed sentiments.
- Embedded and clustered statements with BGE-Large; graph analysis surfaced cross-disciplinary research gaps and ranked hypotheses; highlighted directions for interdisciplinary improvement.
- Made *ZetaEvolve*, an AlphaEvolve-inspired genetic-algorithm engine that turns hypotheses into fast, testable "microexperiments".

Honors & Awards

Regeneron ISEF Grand Award: 1st Place Biomedical Engineering (2024)

Optiver Opticode Cipher Challenge at NeurIPS: First Place Winner (2025)

UC Berkeley Citadel Securities Trading Challenge: First Place Winner (2025)

MIT CRE[AT]E Assistive Technology Competition: Winner (2024)

Pfizer DRL Global AI Flight Competition Finalist: Top 3 contestants internationally (Nominated for Grand Prize) (2025)

Stanford BASES Challenge: Round 2 Qualifier (top 20 of 200+ startups) (2025)

U.S. National Gallery for Young Inventors: 1 of 6 selected nationwide (2024)

UC COSMOS: Accepted into the AI Cluster at UCSC's COSMOS program (∼ 5% accepted) (2023)

NSPA Specialty Magazines Best Of Show: Ranked 6th nationally for *Veritas* (2023)

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, Swift

ML/AI: PyTorch, TensorFlow, CUDA, Transformers, YOLO, Weights & Biases

Tools: Docker, Git, Linux, Pandas, Scikit-Learn, Xcode, OpenCV, Conda, React, Next.js, Node.js, Vercel, Solidity, Noir

Hobby: Classically trained tenor singer, v5 rock climber @ Berkeley Mosaic, film & digital photography, print & digital journalism